Day-31

IPV6 Part (I)

What about IPv5 ?

↳ 'Internet Stream Protocol' was developed in late 1970s, but never actually introduced for public use.
It was never called 'IPv5'; but it used a value of 5 in the Version field of IP header.
So when successor to IPv5 IPv4 was being developed, it was named IPv6. It uses ver value of 6 in version field header.

Number Systems to learn

- Binary / Base 2 / 0b
  0, 1 0b10 = 0d02 (Binary में 10 = decimal में 2)

- Decimal / Base 10 / 0d
  0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

- Hexadecimal / Base 16 / 0x
  0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

Some conversions.

0b1101011011 ko into hexadecimal.

↳ Each hex digit contains 4 bits of information

0b1101 1011

0d13=D 0d11

0xD 0xB

So, 0b1101011011 = 0x3DB

4 · 0b00101111 = 0x2F

Hexadecimal to Binary

0xEC = 0b11101100

0xE = ~~bad~~ E = 14 → 0b1110

0xC0 → C = 12 → 0b1100

So, 0xEC = ~~0b0~~ 0b11101100

Why IPv6. ?

- The main reason for IPV6 is that there ~~aren't~~ simply aren't enough IPv4 address available
- There are 4,294,967,296 (2^{32}) IPv4 address available.
- When IPv4 was being designed 30 years ago, the creators had no idea, the Internet would be as large as it is today.
- VLSM, private IPv4 addresses and NAT have been used to conserve the use of IPv4 address space.

or

- These are short term solutions.
- The long term solution is IPv6

- IPv4 address assignments are controlled by IANA (Internet Assigned Numbers Authority).

- IANA distributes IPv4 address space to various RIRs (Regional Internet Registries), which then assign them to companies that need them.

{ Many of the regions had announced that they made their final IPv4 allocation & they don't have anymore }

Scanned with CamScanner

IPv6

- IPv6 is 128 bits
- So, no. of possible address = 2^{128}
- Its not 4 * the no. of IPv4 address as every additional bit doubles the no. of possible addresses

- If we write IPv6 in bits 128 digits, so even if we make octet its 16 octets
- So, IPv6 are denoted using hexadecimal (Remember one hex digit = 4 bits)
  So, th IPv6 address is represented by 32 hex digits

- & 4 hex digits are grouped together & separated with colons(:).
  So, the are 8 group each containing 4 hex digits. (8x4 = 32 hex)
  Example:

2001:0DB8:5917:EABD:6562:17EA:C92D:59BD /64

IPv6 addresses are longer & more difficult to read & remember, than IPv4 address, but that's unavoidable

- IPv6 addresses use the 'slash' notation to indicate the prefix length, (even when configuring address in CISCO IOS all) Finally, no more dotted decimal subnet mask (like 255.255.255.0 instead of /24)

Shortening (abbreviating IPv6 address)

- Leading 0s can be removed

2001:0DB8:000A:001B:20A1:0020:0080:34BD

↓

Shortened one:

2001:DB8:A:1B:20A1:20:80:34BD

- Consecutive quartets of all 0s can be replaced by a ~~semi~~ double colon (::)

2001:0DB8:0000:0000:0000:0000:0080:34BD

↓

2001:0DB8::0080:34BD

We know IPv6 address has 8 quartets. We can only see 4 quartets. So, we know ~~a~~ double colon means that there are 4 quartets of all 0's.

2001:0DB8::0080:34BD

↓ ↓

~~2001:DB8::80::34~~ 2001:DB8::80:34BD

- Consecutive quartes of 0s can be only be abbreviated once ~~in~~ in IPv6 address.

2001:0000:0000:0000:20A1:0000:0000:34BD

↓

2001::20A1::34BD

How many quartets of 0s are here?

How many quartes of 0s are Here,

So, we do 2001::20A1:0:0:34BD

Expanding shortened IPv6 address

- Put leading 0s where needed (all quartets should have 4 hexadecimal characters).

FE80::2:0:0:FBE8

↓

FE80::0002:0000:0000:FBE8 (5 quartets only)

- If a double colon is used, replace it with all 0-quartets. Make sure there are 8 quartets in total.

↓

FE80:0000:0000:0000:0002:0000:0000:FBE8

Finding the IPv6 prefix (Global Unicast address)

{How to find IPv6 prefix, the network address, from a particular host address}

- Typically, an enterprise requesting IPv6 addresses from their ISP will receive a /48 block.

- Typically, IPv6 subnets use a /64 prefix length.
- This means an enterprise has 64 bits to use to make subnets.
- The remaining 64 bits can be used for hosts

{Global routing prefix}

2001:0DB8:8B00:0007 : 0000 : 0000:0000:0001 /64

- 48-bit 'global routing prefix' assigned by the ISP
- 16 bit subnet identifier used by enterprise to make various subnet
- 64-bit interface identifier, the host portion of address

Scanned with CamScanner

Finding the prefix of an IPv6 address ~~with /64~~

2001:0DB8:8B00:0001:0000:0000:0000:0001 /64

↓

Finding the prefix of an IPv6 address with /64 as prefix length is easy. Simply make the second half of the address all 0s.

2001:DB8:8B00:1::/64

{Here we've replaced host portion with ie calls 0s with (::) double-colon}

- Even if prefix length isn't /64, if the prefix length is multiple of 4 its easy to find a prefix length. It's because each hex character is 4 bits)

Example.

300D:00F2:0B34:2100:0000:0000:1200:0001 /56

↓

300D:F2:B34:2100::/56

{Find where network address portion ends & change remaining parts to 0s}

- 2001:0DB8:8B00:0001:FB89:017B:0020:0011 /93

16 32 48 64 80 84 88 92

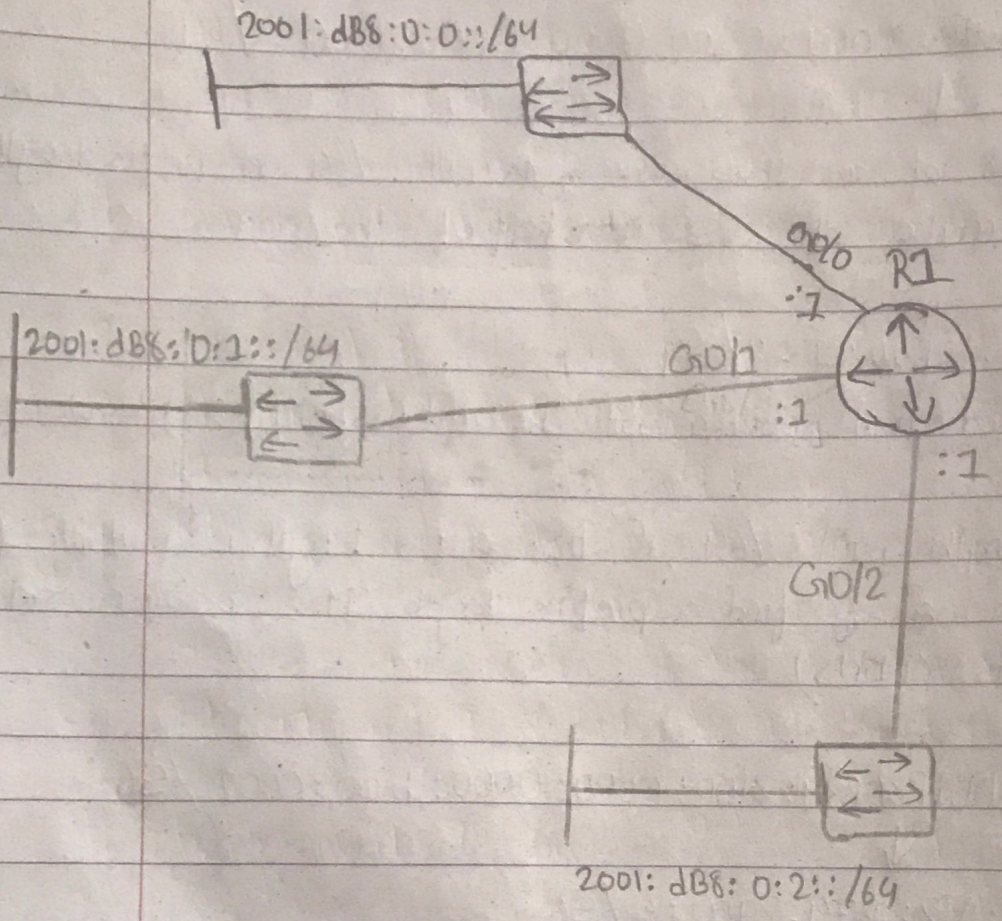
B = 0d11 = 0b1011 = 0b1000 {Changing remaining bits to 0s as they are host portion}

0b1000 = 0d8 → 0x8

2001:DB8:8B00:1:FB89:178::/93

{2001:DB8 has been reserved for examples & documentation}

Configuring IPv6 address.



R1(config)# ipv6 unicast-routing.
R1 (config)# int g0/0
R1(config-if)# ipv6 add 2001:dB8:0:0::1/64
R1 (config-if)# no shutdaon.
R1(config-if)# do show ipv6 interface brief.